

Singular Basins in Multiscale Systems: Tunneling between Stable States

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Real-world systems often evolve on different timescales and possess multiple coexisting stable states. Whether or not a system returns to a given stable state after being perturbed away from it depends on the shape and extent of its basin of attraction. We show that basins of attraction in multiscale systems can exhibit special geometric properties in the form of *singular funnels*. Although singular funnels are narrow, they can extend to different regions of the phase space and, unexpectedly, impact the system's resilience to perturbations. Consequently, singular funnels may prevent common dimensionality reductions in the limit of large timescale separation, such as the quasistatic approximation, adiabatic elimination, and time averaging of the fast variables. We refer to basins of attraction with singular funnels as *singular basins*. We show that singular basins are universal and occur robustly in a range of multiscale systems: the normal form of a pitchfork bifurcation with a slowly adapting parameter, an adaptive active rotator, and an adaptive network of phase rotators.

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Complex real-world systems are characterized by *multistability*; i.e., when started from different initial conditions, the same system can end up with notably different asymptotic behavior. Examples of multistability can be found in epileptic and neuronal models [1,2], reservoir computers [3], lasers [4,5], and climate [6,7], to name just a few. The analysis and control of multistability-related phenomena has therefore been the subject of many studies [8–10]. An important question in multistable systems concerns possible transitions between different stable states. One notable approach addressing the above challenges is the concept of basin stability [11,12], which represents the likelihood of reaching a given state (attractor) when starting from a random initial condition. This likelihood can be estimated by the relative volume of its basin of attraction, i.e., the proportion of initial states in the phase space that converge to this attractor. More importantly, the shape of the basins of attraction determines the uncertainty [13] and resilience [14,15] of a system to perturbations. Therefore, to understand and control a multistable system, it is crucial to understand the geometry of its basins of attraction.

In addition to having multiple stable states, it is also common for real-world systems to evolve on *multiple timescales*, which gives rise to a variety of nonlinear phenomena [16–20]. An important example of a multiscale system is an adaptive dynamical network, where the adaptation is much slower than the node dynamics [21–26]. Such systems are challenging because they are also high dimensional and can exhibit a high degree of multistability. The study of basins of attraction in multiscale systems is an area of research that remains largely unexplored. Therefore, uncovering new properties of the basins that arise from the interplay of different timescales is an intriguing and important task.

Useful and widely used techniques have been developed to analyze mathematical models of multiscale systems with two distinct timescales, also known as slow-fast systems [27–29]. These techniques exploit the limit of infinite ratio between the fast and slow timescales and can be broadly divided into two groups [16,30,31]. The first group includes adiabatic elimination and averaging. It eliminates the fast variables by treating them as instantaneous, or by taking into account their effective contribution, while only considering the time evolution of the slow variables in the *reduced system*. The second group includes the quasistatic approximation. It “freezes” the slow variables by treating them as static at each instant, while only considering the time evolution of the fast variables in the *layer system* [16,28].

This Letter uncovers special properties of basins of attraction in slow-fast systems, leading to counterintuitive

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phenomena. We show that such basins can contain *singular funnels* (SFs) in the form of tunnels that extend to different regions of the phase space and become increasingly narrow as the timescale ratio increases. The problem with SFs is that they are eliminated by the quasistatic approximation and adiabatic elimination. Consequently, adiabatic elimination makes it impossible to reach a given stable state from certain regions of the phase space, whereas this is possible in the full system. On the other hand, the quasistatic approximation could mask the presence of multiple basins of attraction entirely. Therefore, one must be cautious when extrapolating the resilience properties of the full system from those of a reduced system, regardless of whether the latter is obtained via quasistatic approximation, adiabatic elimination, or averaging. Because of the vanishing of SFs in the limit of infinite ratio of the timescales, we refer to basins containing SFs as *singular basins*.

We begin by illustrating the surprising effects of SFs in the normal form for a supercritical pitchfork bifurcation with a slowly adapting bifurcation parameter. The normal form alone is given by $dx/dt = x(\mu - x^2)$ [32]. We restrict to $x \geq 0$, which is appropriate for many applications, for example, when the variable x denotes a population density or the amplitude of a certain observable. Such system has a single stable equilibrium x^* , which exists continuously for all values of μ . Specifically, $x^* = 0$ for $\mu \leq 0$ and $x^* = \sqrt{\mu}$ for $\mu > 0$, so it can be written in a more compact form as $x^*(\mu) = \sqrt{\mu H(\mu)}$, where $H(\mu)$ is the Heaviside step function [33]. We then introduce the slow variable in the form of a linear adaptation of the bifurcation parameter μ , resulting in the following slow-fast system written in terms of the *fast time* t :

$$\frac{dx}{dt} = x(\mu - x^2), \quad (1)$$

$$\frac{d\mu}{dt} = \varepsilon(-\mu + ax - b). \quad (2)$$

The small parameter $0 < \varepsilon \ll 1$ is the ratio between the slow timescale of μ and the fast timescale of x . When $\varepsilon = 0$, the slow timescale is infinitely slower, or equivalently, the fast timescale is infinitely faster. The parameters $a, b > 0$ determine the adaptation rule. Under the condition $a > 2\sqrt{b}$, system (1)–(2) has two stable equilibria, e_1 and e_3 , one saddle equilibrium, e_2 , and an SF, as shown in Fig. 1(a); see Ref. [34] for more details.

The layer system is obtained by setting $\varepsilon = 0$ in (1)–(2), which gives system (1) with a static μ . It has a unique stable quasistatic equilibrium $x^*(\mu)$, which varies continuously with μ and attracts all initial conditions $x_0 > 0$ at every value of μ [the solid part of the gray curve S in Fig. 1(a)]. Therefore, we apply the adiabatic elimination procedure. We therefore substitute $x^*(\mu)$ into (2), define the *slow time* to be $\tau = \varepsilon t$, and obtain the reduced system in terms of τ ,

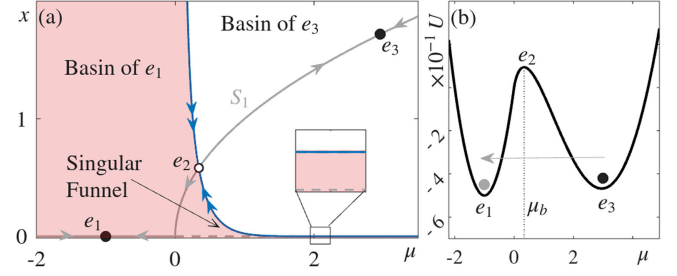


FIG. 1. (a) Singular basin with a singular funnel (SF) in a bistable pitchfork normal form system (1)–(2) with slowly adapting parameter μ shown on the horizontal axis. The red (white) region is the basin of attraction of the stable equilibrium e_1 (e_3). The boundary of the two basins is given by the stable manifold of the saddle equilibrium e_2 (blue curves). The SF near $x = 0$ shrinks exponentially with μ . (b) The bistable potential of the corresponding slow subsystem after adiabatic elimination of the fast variable x (equivalently, reduction to the stable critical manifold). Parameters are $a = 3$, $b = 2$, and $\varepsilon = 0.1$.

$$\frac{d\mu}{d\tau} = f(\mu), \quad (3)$$

where $f(\mu) = -\mu + a\sqrt{\mu H(\mu)} - b$. We then recognize that this is a bistable dynamical system with a double-well potential,

$$U(\mu) = - \int f(\mu) d\mu = \mu^2/2 + b\mu - 2a[\mu H(\mu)]^{3/2}/3 \quad (10)$$

and the same equilibria as the full system; see Fig. 1(b). After adiabatic elimination, the basins of attraction of the stable equilibria e_1 and e_3 in Fig. 1(b) are clearly separated by a single value of the slow variable, $\mu = \mu_b$, given by the position of the unstable equilibrium e_2 . If the reduced system is started on the left (right) of μ_b , it can only converge to e_1 (e_3). However, there is a surprise. This clear separation of basins by $\mu = \mu_b$ is never present in the full system (1)–(2) due to the SF, no matter how large the timescale ratio is (or how small but nonzero the value of ε is). In particular, the SF allows convergence to e_1 from both sides of μ_b , which is impossible in the reduced system. This includes perturbations from e_3 into the SF along the fast x direction. From the reduced system's perspective, this appears as though there is quantumlike “tunneling” or “teleporting” from e_3 to e_1 [the gray arrow in Fig. 1(b)]. In summary, the SF clearly restricts the applicability of the adiabatic elimination, even though the layer system has a unique stable quasistatic equilibrium which varies continuously with μ and attracts all initial conditions $x_0 > 0$ at every value of μ . To demonstrate that the failure of a global adiabatic elimination for system (1)–(2) is not caused by the degeneracy at $(\mu, x) = (0, 0)$, Supplemental Material [34] presents a modified system that does not have such degeneracy, but still has an SF.

The SF in the paradigmatic model (1)–(2) has a very simple geometry. In general, however, SFs can have intricate geometries and can penetrate the (white) basin of the other stable state in complicated ways. We will now demonstrate this for adaptive phase rotators.

The fast-slow adaptive phase rotator has the form [35]

$$\frac{d\varphi}{dt} = \omega + \mu - \sin \varphi, \quad (4)$$

$$\frac{d\mu}{dt} = \varepsilon \{-\mu + \eta[1 - \sin(\varphi + \alpha)]\}, \quad (5)$$

where $\varphi \in (0, 2\pi]$ is a phase variable, ω is the base oscillator frequency, and μ accounts for a slow self-adjustment of the oscillator frequency. The layer system (4) with static μ has a pair of equilibria, one attracting and one repelling, for $|\omega + \mu| < 1$, and a periodic rotation for $|\omega + \mu| > 1$ [36]. At the point $|\omega + \mu| = 1$, the layer system undergoes a saddle node on invariant circle bifurcation [37–40]. Hence, the layer system (4) has a unique attracting quasistatic equilibrium, which varies continuously with μ and attracts all initial conditions (except for the repelling equilibrium point) at every value of μ , when $|\omega + \mu| \leq 1$. The attracting equilibrium is replaced by a periodic rotation that exists when $|\omega + \mu| > 1$.

The adaptive phase rotator (4)–(5) can also be reduced to the form (3) by eliminating the fast variable φ . This is achieved by averaging over the fast rotations for $|\omega + \mu| > 1$, and by adiabatic elimination for $|\omega + \mu| < 1$. The corresponding function $f(\mu)$ of the reduced system (3) can be calculated explicitly [34]:

$$f(\mu) = -\mu + \eta[1 - (\mu + \omega) \cos \alpha + \Omega(\mu)], \quad (6)$$

where

$$\Omega(\mu) = \begin{cases} -\sqrt{1 - (\mu + \omega)^2} \sin \alpha & \text{for } |\mu + \omega| \leq 1 \\ \sqrt{(\mu + \omega)^2 - 1} \cos \alpha & \text{for } \mu + \omega > 1 \\ -\sqrt{(\mu + \omega)^2 - 1} \cos \alpha & \text{for } \mu + \omega < -1. \end{cases}$$

We consider the case when system (4)–(5) possesses two coexisting attractors: a stable equilibrium e_1 and a stable rotation γ_c ; see Fig. 2(a) and [34] for more details. The reduced system (3), with $f(\mu)$ given by Eq. (6), is also a bistable system with a double-well potential as shown in Fig. 2(b).

The singular basin of the adaptive phase oscillator [system (4)–(5)] in Figs. 2(a) and 2(c) has an SF that extends into the region of negative μ . This creates a channel through which initial conditions with arbitrary μ and selected φ are attracted to the periodic rotation γ_c . Once again, this is impossible within the framework of the reduced system (3), where only initial conditions with

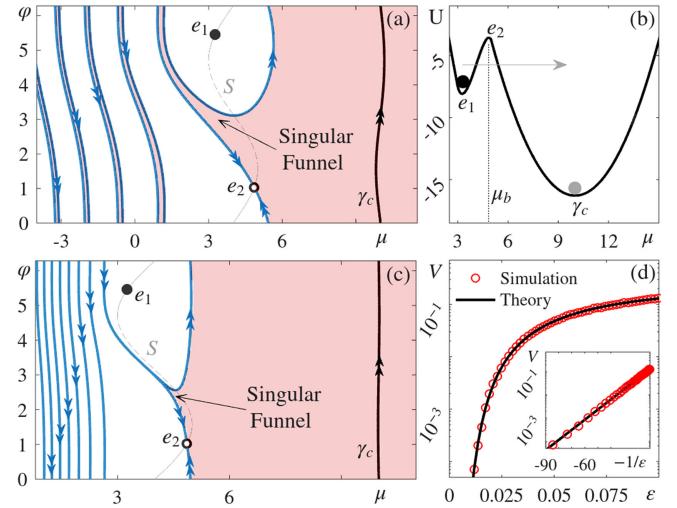


FIG. 2. (a),(c) Singular basin with an SF in an adaptive phase oscillator system (4)–(5). The basin of attraction of the periodic rotation γ_c is plotted in red and the basin of attraction of the stable equilibrium e_1 is plotted in white. The (blue) boundary of the two basins is given by the stable manifold of the saddle equilibrium e_2 . (b) The bistable potential of the corresponding slow subsystem after adiabatic elimination and averaging of the fast variable φ . (d) The volume of the SF limited to $-10 \leq \mu \leq 3$ as a function of ε obtained using Eq. (7) (black curve) and Monte Carlo simulation (red open circles) [34]. Other parameters are $\eta = 10$, $\omega = -4$; (a) $\alpha = \pi/2$, $\varepsilon = 0.1$; (c) $\alpha = \pi/2$, $\varepsilon = 0.01$.

$\mu > \mu_b$ can converge to the averaged periodic rotation γ_c ; see Fig. 2(b).

We emphasize here that the geometry of the SF of the adaptive phase oscillator [system (4)–(5)] is more complex than that of the adaptive pitchfork normal form (1)–(2). The main differences are the following. (i) The SF extends over the entire range of the phase variable φ in Fig. 2, as opposed to an exponentially small region of the x variable in Fig. 1. (ii) The SF in Fig. 1 decreases exponentially with μ , with μ increasing up to infinity, whereas in Fig. 2, such an exponential decrease only occurs over a finite μ interval. This leads to a higher volume of the SF for the system (4)–(5) compared to the system (1)–(2). To provide a better insight into the geometry of the SF for system (4)–(5), we have included the phase portrait with the variable φ shown on the real line in Supplemental Material [34].

Next, we provide details of how the boundaries of the SF in Figs. 2(a) and 2(c) are formed. As in the case of the adaptive pitchfork normal form system (1)–(2), these boundaries consist of orbits attracted to the saddle equilibrium e_2 (branches of the stable invariant manifold of e_2). Tracking these orbits backward in time reveals that they pass close to the branch of unstable quasistatic equilibria of the layer system. This is the dashed part of the curve S in Figs. 2(a) and 2(c), where S is also known as the critical manifold [16]. Since this part of the critical manifold is attractive in backward time, the two SF boundaries

exponentially approach each other as they are attracted to S , causing the SF to become exponentially narrow. After the SF extends past S to negative μ , it undergoes further rotations, repeatedly intersecting the white basin of attraction of e_1 in a complicated pattern.

For the systems with a singular basin that have been considered so far, we can obtain the universal scaling of how the SF volume shrinks as ε approaches 0. In all cases, the motion along the trajectories which correspond to the SF boundaries is slow and its speed is proportional to ε along the unstable branch of a critical manifold S . Thus, the time they spend close to an unstable branch of S of length L scales as L/ε . The width δ of the SF is determined by two trajectories that pass close to S (blue trajectories in Fig. 2). Hence, δ decreases exponentially as time goes backward. Therefore, this width can be estimated as $\delta \sim \exp(-L\lambda/\varepsilon)$ at the moment when the SF leaves the neighbourhood of an unstable branch of S of length L . Here λ is an effective repulsion rate away from S , which corresponds to the attraction in reverse time. These arguments lead to the following expected scaling for the volume of an SF:

$$V(\varepsilon) \sim \exp(-C\varepsilon^{-1}), \quad C > 0. \quad (7)$$

The scaling (7) holds for the adaptive normal form system (1)–(2), since the stable manifold of e_2 is exponentially repelled from $x = 0$. Furthermore, the adaptive phase rotator system (4)–(5) exhibits this scaling too, as shown in Fig. 2(c). The SF scaling (7) is expected to hold for large classes of slow-fast systems, at least in low dimensions, as the arguments leading to (7) are based on the rather general geometric structure of the basin boundaries. However, as we show below, SFs with even more intricate geometries found in higher-dimensional systems may show deviations from this scaling.

A higher-dimensional class of systems, in which singular basins can be observed, is the system of mean-field coupled active rotators,

$$\frac{d\varphi_i}{dt} = \omega_i + \mu - \sin \varphi_i + \frac{\kappa}{N} \sum_{j=1}^N \sin(\varphi_j - \varphi_i), \quad (8)$$

$$\frac{d\mu}{dt} = \varepsilon[-\mu + \eta(1 - X)], \quad (9)$$

where $i = 1, \dots, N$ and $X = (1/N) \sum_{j=1}^N \sin(\varphi_j + \alpha)$. In this system, the frequencies of the individual rotators are adapted globally by the slow variable μ , and μ is driven by the mean field X . In contrast to the extensively studied system of coupled rotators without adaptation [10,41–47], the adaptive system (8)–(9) evolves on two distinct time-scales with ratio ε . Adaptive systems of a similar nature have been shown to exhibit distinct dynamical properties such as canard cascading [18,48], emergent excitability

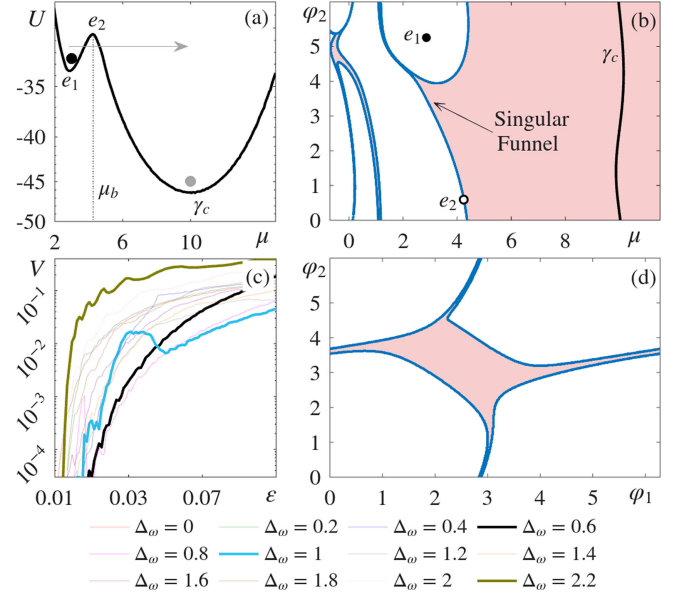


FIG. 3. (a) Bistable potential of the slow subsystem of two adaptive phase oscillators (8)–(9), for $N = 2$, after adiabatic elimination and averaging of the fast variables $\varphi_{1,2}$. (b),(d) Two cross sections of the singular basin with an SF of the same system: (b) $\varphi_1 = 1.2461$ and (d) $\mu = 2.86$. The basin of attraction of the periodic rotation γ_c is plotted in red and the basin of attraction of the stable equilibrium e_1 is plotted in white. The (blue) boundary of the two basins is given by the stable manifold of the saddle equilibrium e_2 . (c) The volume of the SF limited to $-10 \leq \mu \leq 0$ as a function of ε obtained using Monte Carlo simulation [34] for different values of the parameter $\Delta_\omega = \omega_1 - \omega_2$, showing the robustness of the SF. Other parameters are $\omega_1 = -4$, $\omega_2 = -3$, $\kappa = 1$, $\eta = 10$, $\alpha = \pi/2$; (b)–(d) $\varepsilon = 0.1$.

[49], and others [50–52]. Here, we add singular basins to this list of distinct properties.

Figure 3 shows a singular basin and its scaling in a system of $N = 2$ rotators. The system features coexistence of two attractors: one stable equilibrium and one periodic or quasiperiodic rotation. A bistable potential calculated by averaging system (8) with $N = 2$ oscillators is shown in Fig. 3(a). Two-dimensional cross sections of the corresponding singular basin for fixed values of φ_1 and μ values are illustrated in Figs. 3(b) and 3(d), respectively.

The following observations shed light on the topological structure of the SF in the three-dimensional model [system (8)–(9)] with $N = 2$. (i) The boundary of the SF is formed by the stable invariant manifold of the saddle point e_2 . Evidence for this can be seen in Fig. 3(b), where the boundary of the basins crosses e_2 (see also Fig. S5 in [34]). (ii) The SF is organized around a nontrivial unstable critical set, which determines the star-shaped cross section of the SF shown in Fig. 3(d); see Sec. V of Supplemental Material [34] for more details.

Figure 3(c) shows how the SF volume shrinks with ε for different values of the detuning parameter $\Delta_\omega = \omega_1 - \omega_2$.

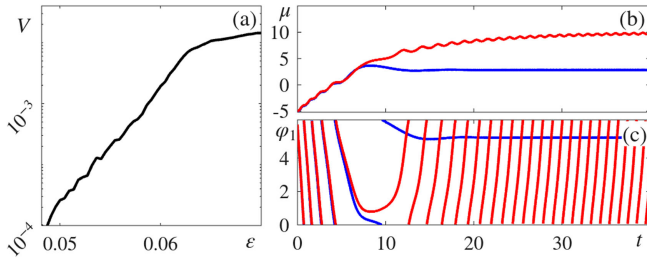


FIG. 4. (a) The volume of the basin of attraction of ten adaptive phase oscillators system (8)–(9) for $N = 10$, limited to $-10 \leq \mu \leq 0$, as a function of ϵ obtained using Monte Carlo simulation [34]. Its decrease with ϵ is consistent with the presence of an SF. (b),(c) Two trajectories starting from close initial conditions: $\phi_i(0) = 6$ for $i = 1, \dots, 10$, with $\mu(0) = -5$ (red) and $\mu(0) = -5.1$ (blue). The red trajectory starts from SF and, therefore, it is attracted to a stable rotating state. Other parameters are $\omega_i = -4 + (i - 1)/9$, $i = 1, \dots, 10$, $\eta = 10$, $\alpha = \pi/2$, and $\kappa = 1$.

We observe that $V(\epsilon)$ follows the scaling (7) for some values of Δ_ω (e.g., $\Delta_\omega = 0.6$), while for other values of Δ_ω , it exhibits resonancelike deviations from (7) (e.g., $\Delta_\omega = 1$ or $\Delta_\omega = 2.2$). Hence, we conclude that higher-dimensional systems can have SFs that deviate from the simple scaling (7) and leave this as a question for future research.

A visualization of SFs in high-dimensional systems via two-dimensional cross sections is very difficult. Instead, the estimated volume of the SF for the case of 10 coupled rotators (11-dimensional phase space) is plotted in Fig. 4(a). Similar to the lower-dimensional examples, the volume shrinks with ϵ , indicating the presence of an SF. Additionally, Fig. 4(b) shows two trajectories: (red) one starting from the SF and converging to periodic rotations, and (blue) one starting and staying very close by, but eventually converging to the stable equilibrium e_1 , which is another indication of an SF.

Finally, singular basins are a robust phenomenon. This is due to the fact that its ingredients persist under small parameter changes. This applies to the stable invariant manifolds of saddle equilibria and the coexisting attractors. We illustrate the wide parameter region in which the singular basin persists and the possible mechanisms by which it is destroyed in system (4)–(5) in [34].

In summary, we have uncovered singular basins with singular funnels in bistable slow-fast systems, and showed that they are robust and ubiquitous. The SFs prevent useful and widely used dimensionality reductions, even when there is a unique quasistatic attractor, which changes continuously with the slow variable(s) and attracts almost all initial conditions at every frozen setting of the slow variable. Although adiabatic elimination or averaging of the fast variables retains the coexistence of different stable states and different basins of attraction in the reduced system, the fast dynamics of the full system can lead to transitions between different stable states via SFs that are

not possible in the reduced system. From the reduced system’s perspective, these unexpected dynamics resemble quantumlike tunneling or teleporting from one attractor to another. We have constructed low-dimensional canonical systems that can be considered normal forms for a singular basin and showed a universal scaling of SFs with increasing timescale ratio in these systems. Additionally, we have demonstrated that SFs can form intricate structures in the phase space of higher-dimensional systems with slowly adaptive mean-field coupling. A challenge for future research is to understand singular basins in systems with more than two timescales and more than two coexisting stable states.

Perhaps most intriguingly of all, any mathematical model of a real-world system is necessarily an approximation that omits some fast variables. Thus, our results also provide new insight into the potential for unexpected critical transitions [53]. For example, transitions between different stable states, which are not captured by the models due to omitting some fast variables, could be triggered in real systems.

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Data availability—The code used to perform the simulations and produce all figures is available in the GitHub repository [54].

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